

Supplementary Material — Part C

Lexical signatures

*What Commentators Talk About: A Validated Thematic Taxonomy of Event-Grounded Sports
Commentary*

The complete set of FDR-significant (cluster, term) pairs from the exact hypergeometric enrichment analysis.

Sections in this part

S5 Full R3 Enriched Vocabulary (864 FDR-Significant Terms)

S5 — Full R3 Enriched Vocabulary (864 FDR-Significant Terms)

This supplement lists all FDR-significant enriched terms for each of the 34 clusters, obtained by an exact one-sided hypergeometric test on binary sentence-level term incidence with a single global Benjamini–Hochberg family ($q \leq 0.05$). Terms are sorted by significance (descending) within each cluster. Over a vocabulary of 3,864 terms and 131,376 candidate (cluster, term) pairs, 864 are FDR-significant across 34 clusters (mean 25.4 per cluster, range 4–115; 54.6% bigrams; no cluster has an empty signature). The mask placeholders are stripped from the text before vectorisation, so <PERSON>, <TEAM> and <PLACE> cannot enter the lexicon while ordinary uses of *person*, *team* and *place* are retained. Columns: **Bi** = bigram indicator; z = significance rank score ($-\log_{10} p$); $\log_2 \text{FC}$ = $\log_2(\text{observed} / \text{expected frequency})$; n = observed count in cluster; **Cov%** = % of cluster sentences containing the term; **Corp%** = expected % under the hypergeometric null (expected / cluster size).

C0 — DRS-assisted battles

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 49

C0 — DRS-assisted battles: 49 FDR-significant terms

Term	Bi	z -score	$\log_2 \text{FC}$	n	Cov%	Corp%
drs		300.00	5.01	202	98.5%	3.1%
got drs	✓	50.80	5.10	34	16.6%	0.5%
drs zone	✓	22.60	5.13	15	7.3%	0.2%
drs got	✓	17.90	5.10	12	5.9%	0.2%
got		15.24	1.67	57	27.8%	8.7%
zone		14.19	3.64	17	8.3%	0.7%
open		12.27	3.74	14	6.8%	0.5%
drs enabled	✓	11.05	5.22	7	3.4%	0.1%
enabled		10.15	5.03	7	3.4%	0.1%
going drs	✓	10.15	5.03	7	3.4%	0.1%
bit drs	✓	9.46	5.22	6	2.9%	0.1%
drs going	✓	9.46	5.22	6	2.9%	0.1%
extra		8.88	3.74	10	4.9%	0.4%
drs open	✓	8.03	4.81	6	2.9%	0.1%
drs time	✓	7.88	5.22	5	2.4%	0.1%
ll drs	✓	7.88	5.22	5	2.4%	0.1%
second drs	✓	7.88	5.22	5	2.4%	0.1%
slipstream drs	✓	7.88	5.22	5	2.4%	0.1%
speed		7.84	2.62	15	7.3%	1.2%
straight		7.25	2.20	18	8.8%	1.9%
helping		5.82	4.37	5	2.4%	0.1%
rear wing	✓	5.66	3.81	6	2.9%	0.2%
200 mph	✓	5.48	3.72	6	2.9%	0.2%
help		5.27	4.08	5	2.4%	0.1%
defensive		5.15	2.94	8	3.9%	0.5%
extra pace	✓	5.14	4.63	4	2.0%	0.1%
slipstream		5.09	2.36	11	5.4%	1.0%
200		5.01	3.48	6	2.9%	0.3%
ll		4.87	1.79	16	7.8%	2.2%
drs getting	✓	4.72	5.23	3	1.5%	0.0%
drs goes	✓	4.72	5.23	3	1.5%	0.0%

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
drs ll	✓	4.72	5.23	3	1.5%	0.0%
drs main	✓	4.72	5.23	3	1.5%	0.0%
drs range	✓	4.72	5.23	3	1.5%	0.0%
drs right	✓	4.72	5.23	3	1.5%	0.0%
drs turn	✓	4.72	5.23	3	1.5%	0.0%
line drs	✓	4.72	5.23	3	1.5%	0.0%
right drs	✓	4.72	5.23	3	1.5%	0.0%
straight drs	✓	4.72	5.23	3	1.5%	0.0%
wing open	✓	4.72	5.23	3	1.5%	0.0%
main straight	✓	4.67	2.98	7	3.4%	0.4%
mph		4.51	3.22	6	2.9%	0.3%
drs wide	✓	4.13	4.81	3	1.5%	0.1%
speed advantage	✓	4.13	4.81	3	1.5%	0.1%
speed drs	✓	4.13	4.81	3	1.5%	0.1%
chance		3.97	1.97	11	5.4%	1.4%
main		3.96	2.64	7	3.4%	0.5%
range		3.74	4.48	3	1.5%	0.1%
advantage		3.64	2.27	8	3.9%	0.8%

C1 — Time penalties

Domain: Regulatory **Verdict:** Strict-core **Enriched terms:** 39

C1 — Time penalties: 39 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
penalty		221.37	5.96	107	90.7%	1.5%
time penalty	✓	80.02	5.98	43	36.4%	0.6%
second penalty	✓	72.10	5.98	39	33.1%	0.5%
second		67.33	3.25	83	70.3%	7.4%
second time	✓	61.04	5.40	41	34.7%	0.8%
10 second	✓	47.70	5.92	27	22.9%	0.4%
causing collision	✓	35.08	6.02	19	16.1%	0.2%
penalty causing	✓	35.08	6.02	19	16.1%	0.2%
collision		34.77	5.77	21	17.8%	0.3%
causing		33.73	5.82	20	16.9%	0.3%
10		28.16	4.21	28	23.7%	1.3%
time		23.71	2.59	45	38.1%	6.3%
penalty forcing	✓	16.44	6.02	9	7.6%	0.1%
got second	✓	15.31	5.53	10	8.5%	0.2%
forcing		15.30	5.23	11	9.3%	0.2%
given second	✓	14.60	6.02	8	6.8%	0.1%
track		14.03	2.96	23	19.5%	2.5%
unsafe release	✓	12.76	6.02	7	5.9%	0.1%
given		12.63	4.10	13	11.0%	0.6%
gaining advantage	✓	12.40	5.56	8	6.8%	0.1%

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
leaving track	✓	12.40	5.56	8	6.8%	0.1%
track gaining	✓	12.40	5.56	8	6.8%	0.1%
release		11.92	5.43	8	6.8%	0.2%
unsafe		11.21	5.65	7	5.9%	0.1%
leaving		11.15	5.21	8	6.8%	0.2%
penalty leaving	✓	10.92	6.01	6	5.1%	0.1%
forcing driver	✓	10.09	5.80	6	5.1%	0.1%
driver track	✓	9.09	6.02	5	4.2%	0.1%
given 10	✓	9.09	6.02	5	4.2%	0.1%
penalty speeding	✓	9.09	6.02	5	4.2%	0.1%
speeding pit	✓	9.09	6.02	5	4.2%	0.1%
advantage		8.71	3.52	11	9.3%	0.8%
speeding		7.78	5.53	5	4.2%	0.1%
gaining		7.59	3.97	8	6.8%	0.4%
got penalty	✓	6.57	5.70	4	3.4%	0.1%
penalty hanging	✓	5.45	6.03	3	2.5%	0.0%
penalty unsafe	✓	5.45	6.03	3	2.5%	0.0%
risk penalty	✓	5.45	6.03	3	2.5%	0.0%
gets second	✓	4.16	5.01	3	2.5%	0.1%

C2 — Lights-out race starts

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 16

C2 — Lights-out race starts: 16 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
lights away	✓	84.41	6.70	41	75.9%	0.7%
lights		78.89	5.92	45	83.3%	1.4%
away		55.42	4.90	41	75.9%	2.5%
away gets	✓	12.19	6.94	6	11.1%	0.1%
away getting	✓	7.96	6.84	4	7.4%	0.1%
start		6.86	2.93	11	20.4%	2.7%
lights lights	✓	6.48	7.16	3	5.6%	0.0%
away ahead	✓	5.88	6.74	3	5.6%	0.1%
start good	✓	5.88	6.74	3	5.6%	0.1%
getting away	✓	4.94	5.94	3	5.6%	0.1%
better start	✓	4.74	5.74	3	5.6%	0.1%
gets away	✓	3.94	4.92	3	5.6%	0.2%
away comes	✓	3.84	6.57	2	3.7%	0.0%
reaction		3.69	3.75	4	7.4%	0.5%
better		3.68	3.18	5	9.3%	1.0%
gets better	✓	3.54	6.16	2	3.7%	0.1%

C3 — Mirror checks defending

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 23

C3 — Mirror checks defending: 23 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
mirrors		47.84	7.16	22	46.8%	0.3%
mirror		36.31	6.93	18	38.3%	0.3%
checking		21.24	6.81	11	23.4%	0.2%
scarlet		16.96	6.71	9	19.1%	0.2%
checks		16.31	7.04	8	17.0%	0.1%
checking mirrors	✓	13.41	7.34	6	12.8%	0.1%
checking left	✓	10.38	7.08	5	10.6%	0.1%
hand mirror	✓	10.38	7.08	5	10.6%	0.1%
left mirror	✓	8.90	7.32	4	8.5%	0.1%
looking mirrors	✓	8.90	7.32	4	8.5%	0.1%
left		8.30	3.62	10	21.3%	1.7%
checks mirrors	✓	6.66	7.38	3	6.4%	0.0%
looking		6.21	3.42	8	17.0%	1.6%
checks mirror	✓	6.06	6.91	3	6.4%	0.1%
driving mirrors	✓	6.06	6.91	3	6.4%	0.1%
left hand	✓	5.58	4.46	5	10.6%	0.5%
sees		5.07	4.89	4	8.5%	0.3%
hand		4.39	3.67	5	10.6%	0.8%
check		4.03	5.03	3	6.4%	0.2%
doesn need	✓	3.96	6.80	2	4.3%	0.0%
having look	✓	3.66	6.32	2	4.3%	0.1%
rival title	✓	3.66	6.32	2	4.3%	0.1%
trying retake	✓	3.66	6.32	2	4.3%	0.1%

C4 — Race start launch

Domain: Race dynamics **Verdict:** Robust **Enriched terms:** 15

C4 — Race start launch: 15 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
launch		82.05	6.87	38	90.5%	0.8%
launch line	✓	41.10	7.24	19	45.2%	0.3%
good launch	✓	26.91	7.13	13	31.0%	0.2%
line		19.95	3.60	23	54.8%	4.5%
good		12.77	3.67	15	35.7%	2.8%
got good	✓	6.60	6.06	4	9.5%	0.1%
got launch	✓	6.21	7.09	3	7.1%	0.1%
gets launch	✓	5.82	6.80	3	7.1%	0.1%
chops		4.90	5.94	3	7.1%	0.1%
pole sitter	✓	4.90	5.94	3	7.1%	0.1%
sitter		4.90	5.94	3	7.1%	0.1%
chop		4.74	5.77	3	7.1%	0.1%
got		4.44	1.83	13	31.0%	8.7%
initial launch	✓	4.06	6.97	2	4.8%	0.0%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
great launch	✓	3.76	6.51	2	4.8%	0.1%

C5 — Weather conditions

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 15

C5 — Weather conditions: 15 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
rain		69.90	7.08	32	74.4%	0.5%
rain coming	✓	8.36	7.16	4	9.3%	0.1%
wet		7.73	5.80	5	11.6%	0.2%
minutes		6.97	6.29	4	9.3%	0.1%
rain falling	✓	6.78	7.46	3	7.0%	0.0%
falling		5.24	6.27	3	7.0%	0.1%
light		4.34	5.36	3	7.0%	0.2%
expected		4.24	5.25	3	7.0%	0.2%
track		4.07	2.70	7	16.3%	2.5%
extraordinary		4.05	5.06	3	7.0%	0.2%
class		4.03	6.88	2	4.7%	0.0%
lasting		4.03	6.88	2	4.7%	0.0%
temperature		3.82	4.81	3	7.0%	0.2%
10 minutes	✓	3.74	6.51	2	4.7%	0.1%
30		3.51	6.16	2	4.7%	0.1%

C6 — Grand Prix wins

Domain: Narrative & meta **Verdict:** Strict-core **Enriched terms:** 115

C6 — Grand Prix wins: 115 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
grand		300.00	3.83	445	95.7%	6.7%
grand prix	✓	300.00	3.83	445	95.7%	6.7%
prix		300.00	3.82	454	97.6%	6.9%
wins		80.58	3.56	88	18.9%	1.6%
wins grand	✓	36.66	3.95	32	6.9%	0.4%
lead grand	✓	32.95	3.99	28	6.0%	0.4%
british		22.18	3.90	20	4.3%	0.3%
british grand	✓	22.18	3.90	20	4.3%	0.3%
win		17.24	2.22	42	9.0%	1.9%
place grand	✓	17.15	3.95	15	3.2%	0.2%
dutch		14.65	4.04	12	2.6%	0.2%
dutch grand	✓	14.65	4.04	12	2.6%	0.2%
prix victory	✓	13.43	4.04	11	2.4%	0.1%
lead		12.89	1.53	57	12.3%	4.3%
prix underway	✓	12.41	3.13	17	3.7%	0.4%

Term	Bi	z-score	$\log_2$ FC	n	Cov%	Corp%
australian		12.37	3.91	11	2.4%	0.2%
australian grand	✓	12.37	3.91	11	2.4%	0.2%
saudi		12.37	3.91	11	2.4%	0.2%
win grand	✓	12.20	4.04	10	2.2%	0.1%
belgian		11.58	3.80	11	2.4%	0.2%
belgian grand	✓	11.58	3.80	11	2.4%	0.2%
italian		11.58	3.80	11	2.4%	0.2%
italian grand	✓	11.58	3.80	11	2.4%	0.2%
lap grand	✓	10.98	4.04	9	1.9%	0.1%
victory		10.82	2.27	25	5.4%	1.1%
grands		10.43	3.78	10	2.2%	0.2%
grands prix	✓	10.43	3.78	10	2.2%	0.2%
underway		10.08	2.65	18	3.9%	0.6%
winner		9.81	2.70	17	3.7%	0.6%
arabian		9.75	4.04	8	1.7%	0.1%
arabian grand	✓	9.75	4.04	8	1.7%	0.1%
prix winner	✓	9.75	4.04	8	1.7%	0.1%
saudi arabian	✓	9.75	4.04	8	1.7%	0.1%
takes		9.64	1.78	33	7.1%	2.1%
victories		9.48	3.41	11	2.4%	0.2%
prix win	✓	8.82	3.87	8	1.7%	0.1%
canadian		8.15	3.72	8	1.7%	0.1%
canadian grand	✓	8.15	3.72	8	1.7%	0.1%
takes lead	✓	8.07	2.68	14	3.0%	0.5%
azerbaijan		7.65	3.85	7	1.5%	0.1%
azerbaijan grand	✓	7.65	3.85	7	1.5%	0.1%
austrian		7.61	3.58	8	1.7%	0.1%
austrian grand	✓	7.61	3.58	8	1.7%	0.1%
start grand	✓	7.61	3.58	8	1.7%	0.1%
formula grand	✓	7.31	4.04	6	1.3%	0.1%
started grand	✓	7.31	4.04	6	1.3%	0.1%
hungarian		7.02	3.68	7	1.5%	0.1%
hungarian grand	✓	7.02	3.68	7	1.5%	0.1%
japanese		7.02	3.68	7	1.5%	0.1%
japanese grand	✓	7.02	3.68	7	1.5%	0.1%
spanish		6.76	3.34	8	1.7%	0.2%
spanish grand	✓	6.76	3.34	8	1.7%	0.2%
row		6.51	2.32	14	3.0%	0.6%
prix going	✓	6.49	3.82	6	1.3%	0.1%
prix year	✓	6.11	3.39	7	1.5%	0.1%
leads grand	✓	6.09	4.04	5	1.1%	0.1%
prix comes	✓	6.09	4.04	5	1.1%	0.1%
end grand	✓	5.45	3.46	6	1.3%	0.1%
abu		5.33	3.78	5	1.1%	0.1%
abu dhabi	✓	5.33	3.78	5	1.1%	0.1%
dhabi		5.33	3.78	5	1.1%	0.1%

Term	Bi	z-score	$\log_2$ FC	n	Cov%	Corp%
dhabi grand	✓	5.33	3.78	5	1.1%	0.1%
prix second	✓	5.33	3.78	5	1.1%	0.1%
prix victories	✓	5.33	3.78	5	1.1%	0.1%
time wins	✓	5.33	3.78	5	1.1%	0.1%
year		5.27	1.62	20	4.3%	1.4%
emilia		4.87	4.04	4	0.9%	0.1%
emilia romagna	✓	4.87	4.04	4	0.9%	0.1%
lead british	✓	4.87	4.04	4	0.9%	0.1%
line wins	✓	4.87	4.04	4	0.9%	0.1%
position grand	✓	4.87	4.04	4	0.9%	0.1%
prix grand	✓	4.87	4.04	4	0.9%	0.1%
prix time	✓	4.87	4.04	4	0.9%	0.1%
retirement grand	✓	4.87	4.04	4	0.9%	0.1%
romagna		4.87	4.04	4	0.9%	0.1%
romagna grand	✓	4.87	4.04	4	0.9%	0.1%
wins abu	✓	4.87	4.04	4	0.9%	0.1%
wins australian	✓	4.87	4.04	4	0.9%	0.1%
wins austrian	✓	4.87	4.04	4	0.9%	0.1%
wins azerbaijan	✓	4.87	4.04	4	0.9%	0.1%
wins belgian	✓	4.87	4.04	4	0.9%	0.1%
wins british	✓	4.87	4.04	4	0.9%	0.1%
won grand	✓	4.87	4.04	4	0.9%	0.1%
second grand	✓	4.81	3.55	5	1.1%	0.1%
ends		4.41	3.36	5	1.1%	0.1%
started		4.31	1.83	13	2.8%	0.8%
chinese		4.19	3.72	4	0.9%	0.1%
chinese grand	✓	4.19	3.72	4	0.9%	0.1%
moment grand	✓	4.19	3.72	4	0.9%	0.1%
prix lights	✓	4.19	3.72	4	0.9%	0.1%
prix row	✓	4.19	3.72	4	0.9%	0.1%
victories row	✓	4.19	3.72	4	0.9%	0.1%
grid		4.16	1.79	13	2.8%	0.8%
final		4.11	1.45	18	3.9%	1.4%
home		4.05	1.55	16	3.4%	1.2%
winning		3.82	2.72	6	1.3%	0.2%
10th grid	✓	3.65	4.04	3	0.6%	0.0%
fifth grand	✓	3.65	4.04	3	0.6%	0.0%
final grand	✓	3.65	4.04	3	0.6%	0.0%
lead saudi	✓	3.65	4.04	3	0.6%	0.0%
leading grand	✓	3.65	4.04	3	0.6%	0.0%
meters grand	✓	3.65	4.04	3	0.6%	0.0%
prix got	✓	3.65	4.04	3	0.6%	0.0%
saudi grand	✓	3.65	4.04	3	0.6%	0.0%
seventh grand	✓	3.65	4.04	3	0.6%	0.0%
stop grand	✓	3.65	4.04	3	0.6%	0.0%
winner australian	✓	3.65	4.04	3	0.6%	0.0%

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
wins canadian	✓	3.65	4.04	3	0.6%	0.0%
wins dutch	✓	3.65	4.04	3	0.6%	0.0%
wins emilia	✓	3.65	4.04	3	0.6%	0.0%
wins italian	✓	3.65	4.04	3	0.6%	0.0%
wins japanese	✓	3.65	4.04	3	0.6%	0.0%
opening		3.64	1.80	11	2.4%	0.7%
retirement		3.63	2.62	6	1.3%	0.2%
leads		3.61	1.42	16	3.4%	1.3%

C7 — Race start lights

Domain: Race dynamics **Verdict:** Moderate **Enriched terms:** 26

C7 — Race start lights: 26 FDR-significant terms

Term	Bi	z-score	log ₂ FC	n	Cov%	Corp%
lights		63.11	6.10	34	94.4%	1.4%
red lights	✓	27.71	7.62	12	33.3%	0.2%
prix underway	✓	27.65	6.63	15	41.7%	0.4%
underway		26.80	6.18	16	44.4%	0.6%
grand		21.04	3.36	25	69.4%	6.7%
grand prix	✓	21.04	3.36	25	69.4%	6.7%
prix		20.81	3.33	25	69.4%	6.9%
red		19.42	6.07	12	33.3%	0.5%
lights time	✓	18.03	7.57	8	22.2%	0.1%
lights away	✓	13.38	5.24	10	27.8%	0.7%
green flag	✓	10.77	5.84	7	19.4%	0.3%
begins		9.67	6.90	5	13.9%	0.1%
racing lights	✓	9.38	7.72	4	11.1%	0.1%
green		8.75	4.95	7	19.4%	0.6%
away		7.90	3.45	10	27.8%	2.5%
come lights	✓	7.54	6.72	4	11.1%	0.1%
away formula	✓	7.02	7.74	3	8.3%	0.0%
begins lights	✓	7.02	7.74	3	8.3%	0.0%
flag		6.89	3.69	8	22.2%	1.7%
racing		6.60	3.56	8	22.2%	1.9%
time grand	✓	5.72	6.74	3	8.3%	0.1%
time		4.27	2.13	10	27.8%	6.3%
begins red	✓	4.19	7.16	2	5.6%	0.0%
lights grand	✓	4.19	7.16	2	5.6%	0.0%
flag come	✓	3.89	6.72	2	5.6%	0.1%
formula goes	✓	3.89	6.72	2	5.6%	0.1%

C8 — Race control flags

Domain: Regulatory **Verdict:** Strict-core **Enriched terms:** 34

C8 — Race control flags: 34 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
flag		178.58	5.62	100	84.7%	1.7%
checkered		42.45	5.72	26	22.0%	0.4%
checkered flag	✓	42.45	5.72	26	22.0%	0.4%
yellow		38.17	5.65	24	20.3%	0.4%
green		33.09	5.08	25	21.2%	0.6%
yellow flag	✓	30.09	5.80	18	15.3%	0.3%
chequered		26.37	5.92	15	12.7%	0.2%
chequered flag	✓	26.37	5.92	15	12.7%	0.2%
green flag	✓	24.87	5.41	17	14.4%	0.3%
red flag	✓	13.37	5.09	10	8.5%	0.2%
red		12.70	4.36	12	10.2%	0.5%
flag flies	✓	10.92	6.01	6	5.1%	0.1%
sees checkered	✓	10.92	6.01	6	5.1%	0.1%
sees		10.81	4.73	9	7.6%	0.3%
nation driver	✓	10.09	3.49	13	11.0%	1.0%
flag sector	✓	10.09	5.80	6	5.1%	0.1%
flies		10.09	5.80	6	5.1%	0.1%
got yellow	✓	10.09	5.80	6	5.1%	0.1%
flags		8.62	5.28	6	5.1%	0.1%
sector		8.22	4.21	8	6.8%	0.4%
nation		8.15	2.84	14	11.9%	1.7%
black		7.27	6.01	4	3.4%	0.1%
flag fly	✓	7.27	6.01	4	3.4%	0.1%
driver		6.24	2.23	15	12.7%	2.7%
fly		6.10	5.43	4	3.4%	0.1%
driver checkered	✓	5.45	6.03	3	2.5%	0.0%
driver look	✓	5.45	6.03	3	2.5%	0.0%
flag flying	✓	5.45	6.03	3	2.5%	0.0%
flag wins	✓	5.45	6.03	3	2.5%	0.0%
sees chequered	✓	5.45	6.03	3	2.5%	0.0%
racing		4.91	2.31	11	9.3%	1.9%
going green	✓	4.85	5.60	3	2.5%	0.1%
waved		4.85	5.60	3	2.5%	0.1%
yellow flags	✓	4.46	5.28	3	2.5%	0.1%

C9 — Formula 1 milestones

Domain: Narrative & meta **Verdict:** Strict-core **Enriched terms:** 36

C9 — Formula 1 milestones: 36 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
formula		227.58	5.38	128	95.5%	2.3%
formula world	✓	21.08	5.63	13	9.7%	0.2%
time formula	✓	16.24	5.10	12	9.0%	0.3%

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
podium formula	✓	14.20	5.54	9	6.7%	0.1%
podium		14.13	2.89	24	17.9%	2.4%
formula career	✓	11.47	5.64	7	5.2%	0.1%
world championship	✓	10.79	3.47	14	10.4%	0.9%
world		10.31	2.82	18	13.4%	1.9%
championship		9.15	2.60	18	13.4%	2.2%
career		8.37	3.92	9	6.7%	0.4%
best formula	✓	7.04	5.84	4	3.0%	0.1%
formula 2025	✓	7.04	5.84	4	3.0%	0.1%
formula podium	✓	7.04	5.84	4	3.0%	0.1%
2025		6.75	4.42	6	4.5%	0.2%
time		6.67	1.62	26	19.4%	6.3%
champion		6.21	2.93	10	7.5%	1.0%
returns		5.88	5.25	4	3.0%	0.1%
personal		5.52	5.02	4	3.0%	0.1%
welcome		5.52	5.02	4	3.0%	0.1%
champion formula	✓	5.28	5.82	3	2.2%	0.0%
formula 2022	✓	5.28	5.82	3	2.2%	0.0%
formula champion	✓	5.28	5.82	3	2.2%	0.0%
formula returns	✓	5.28	5.82	3	2.2%	0.0%
welcome formula	✓	5.28	5.82	3	2.2%	0.0%
season		5.07	2.36	11	8.2%	1.6%
victory		4.80	2.60	9	6.7%	1.1%
win formula	✓	4.68	5.42	3	2.2%	0.1%
personal best	✓	4.29	5.09	3	2.2%	0.1%
sport		4.29	5.09	3	2.2%	0.1%
win		4.28	2.08	11	8.2%	1.9%
time year	✓	3.99	4.84	3	2.2%	0.1%
victory formula	✓	3.99	4.84	3	2.2%	0.1%
2023		3.76	4.61	3	2.2%	0.1%
stand		3.75	3.75	4	3.0%	0.2%
driving		3.74	3.20	5	3.7%	0.4%
best		3.57	2.45	7	5.2%	1.0%

C10 — Chicane moves

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 8

C10 — Chicane moves: 8 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
chicane		123.42	6.35	60	96.8%	1.2%
second chicane	✓	13.24	6.58	7	11.3%	0.1%
going chicane	✓	11.81	6.72	6	9.7%	0.1%
goes chicane	✓	6.29	6.97	3	4.8%	0.0%
braking		4.26	3.14	6	9.7%	1.1%
bold		3.97	4.95	3	4.8%	0.2%

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
chicane going	✓	3.71	6.38	2	3.2%	0.0%
make chicane	✓	3.71	6.38	2	3.2%	0.0%

C11 — Team radio celebrations

Domain: Narrative & meta **Verdict:** Strict-core **Enriched terms:** 36

C11 — Team radio celebrations: 36 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
thank		34.93	6.09	21	38.2%	0.6%
job		21.25	5.76	14	25.5%	0.5%
mate		20.25	5.87	13	23.6%	0.4%
weekend		19.30	5.09	15	27.3%	0.8%
guys		19.03	5.31	14	25.5%	0.6%
amazing		11.55	5.53	8	14.5%	0.3%
fantastic		11.38	6.02	7	12.7%	0.2%
thanks		10.50	5.68	7	12.7%	0.2%
unbelievable		10.50	5.68	7	12.7%	0.2%
great job	✓	7.45	6.54	4	7.3%	0.1%
thank guys	✓	7.45	6.54	4	7.3%	0.1%
drive		7.40	3.93	8	14.5%	1.0%
good job	✓	7.08	6.32	4	7.3%	0.1%
yeah		6.91	2.60	13	23.6%	3.9%
awesome		6.45	7.09	3	5.5%	0.0%
deserve		6.45	7.09	3	5.5%	0.0%
job guys	✓	6.45	7.09	3	5.5%	0.0%
job mate	✓	6.45	7.09	3	5.5%	0.0%
weekend thank	✓	6.45	7.09	3	5.5%	0.0%
great		6.45	3.23	9	16.4%	1.7%
love		6.12	4.80	5	9.1%	0.3%
nice		5.83	4.03	6	10.9%	0.7%
good		5.62	2.69	10	18.2%	2.8%
world champions	✓	4.92	5.91	3	5.5%	0.1%
yes		4.74	3.42	6	10.9%	1.0%
difficult		4.64	4.53	4	7.3%	0.3%
beautiful		4.39	5.38	3	5.5%	0.1%
really really	✓	4.39	5.38	3	5.5%	0.1%
champions		4.02	5.00	3	5.5%	0.2%
year		3.97	2.96	6	10.9%	1.4%
really good	✓	3.92	4.89	3	5.5%	0.2%
really		3.87	2.60	7	12.7%	2.1%
congrats		3.82	6.51	2	3.6%	0.0%
executed		3.82	6.51	2	3.6%	0.0%
good weekend	✓	3.82	6.51	2	3.6%	0.0%
yes yes	✓	3.82	6.51	2	3.6%	0.0%

C12 — Podium finishes**Domain:** Narrative & meta **Verdict:** Robust **Enriched terms:** 36

C12 — Podium finishes: 36 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
podium		213.72	5.24	127	91.4%	2.4%
podium places	✓	21.20	5.42	14	10.1%	0.2%
home		18.75	3.75	22	15.8%	1.2%
place podium	✓	13.06	5.61	8	5.8%	0.1%
comes home	✓	10.55	4.30	10	7.2%	0.4%
podium place	✓	10.49	5.78	6	4.3%	0.1%
completes		10.25	4.87	8	5.8%	0.2%
places		10.00	3.13	15	10.8%	1.2%
finish		9.94	3.64	12	8.6%	0.7%
completes podium	✓	9.76	5.13	7	5.0%	0.1%
podium finish	✓	9.06	5.37	6	4.3%	0.1%
complete podium	✓	7.96	5.52	5	3.6%	0.1%
step		7.63	4.34	7	5.0%	0.2%
step podium	✓	7.56	4.78	6	4.3%	0.2%
stand podium	✓	7.42	5.30	5	3.6%	0.1%
come home	✓	7.30	4.67	6	4.3%	0.2%
podium season	✓	6.29	5.46	4	2.9%	0.1%
rostrum		6.29	5.46	4	2.9%	0.1%
complete		5.72	3.91	6	4.3%	0.3%
career podium	✓	5.23	5.77	3	2.2%	0.0%
double podium	✓	5.23	5.77	3	2.2%	0.0%
final podium	✓	5.23	5.77	3	2.2%	0.0%
gonna podium	✓	5.23	5.77	3	2.2%	0.0%
seventh podium	✓	5.23	5.77	3	2.2%	0.0%
home second	✓	5.16	4.10	5	3.6%	0.2%
stand		5.02	4.02	5	3.6%	0.2%
season		4.92	2.31	11	7.9%	1.6%
time		4.75	1.39	23	16.5%	6.3%
home podium	✓	4.63	5.36	3	2.2%	0.1%
career		4.54	3.28	6	4.3%	0.4%
campaign		3.95	4.78	3	2.2%	0.1%
finish season	✓	3.95	4.78	3	2.2%	0.1%
final		3.85	2.20	9	6.5%	1.4%
home finish	✓	3.71	4.56	3	2.2%	0.1%
return		3.51	4.37	3	2.2%	0.1%
place		3.51	1.09	24	17.3%	8.1%

C13 — Championship & title**Domain:** Narrative & meta **Verdict:** Robust **Enriched terms:** 36

C13 — Championship & title: 36 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
championship		84.59	4.40	74	46.8%	2.2%
world		78.05	4.48	67	42.4%	1.9%
world championship	✓	36.87	4.47	33	20.9%	0.9%
champion		36.12	4.41	33	20.9%	1.0%
world champion	✓	19.44	4.23	19	12.0%	0.6%
title		14.78	4.11	15	9.5%	0.5%
constructors		13.78	4.54	12	7.6%	0.3%
win		13.68	2.91	23	14.6%	1.9%
time world	✓	11.78	4.60	10	6.3%	0.3%
championship leader	✓	11.25	4.46	10	6.3%	0.3%
trophy		9.79	5.08	7	4.4%	0.1%
2025		9.51	4.60	8	5.1%	0.2%
points		8.23	2.72	15	9.5%	1.4%
leader		7.69	3.20	11	7.0%	0.8%
2024		7.52	4.72	6	3.8%	0.1%
seven time	✓	7.23	4.60	6	3.8%	0.2%
champions		6.97	4.48	6	3.8%	0.2%
time champion	✓	6.76	5.59	4	2.5%	0.1%
world title	✓	6.76	5.59	4	2.5%	0.1%
championship battle	✓	6.38	4.75	5	3.2%	0.1%
champion world	✓	6.06	5.28	4	2.5%	0.1%
championship lead	✓	5.83	4.46	5	3.2%	0.1%
constructors championship	✓	5.60	4.33	5	3.2%	0.2%
constructors title	✓	5.06	5.60	3	1.9%	0.0%
gap world	✓	5.06	5.60	3	1.9%	0.0%
2022		5.05	4.01	5	3.2%	0.2%
seven		4.48	2.91	7	4.4%	0.6%
constructors champions	✓	4.47	5.18	3	1.9%	0.1%
tally		4.47	5.18	3	1.9%	0.1%
win world	✓	4.47	5.18	3	1.9%	0.1%
time		4.30	1.26	24	15.2%	6.3%
today		4.29	2.57	8	5.1%	0.9%
championship points	✓	4.08	4.86	3	1.9%	0.1%
score		3.96	3.89	4	2.5%	0.2%
contenders		3.70	3.69	4	2.5%	0.2%
winner		3.64	2.76	6	3.8%	0.6%

C14 — Pit stops

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 41

C14 — Pit stops: 41 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
pit		188.89	4.61	139	75.1%	3.1%
pit stop	✓	58.45	4.52	50	27.0%	1.2%
lane		58.41	4.70	47	25.4%	1.0%
pit lane	✓	57.04	4.75	45	24.3%	0.9%
stop		47.71	3.36	63	34.1%	3.3%
pits		38.81	3.97	41	22.2%	1.4%
coming pit	✓	11.36	5.37	7	3.8%	0.1%
comes pit	✓	8.30	4.95	6	3.2%	0.1%
come pits	✓	6.56	4.25	6	3.2%	0.2%
comes		6.47	1.73	23	12.4%	3.7%
coming		6.28	1.99	18	9.7%	2.4%
come pit	✓	5.79	5.05	4	2.2%	0.1%
pitted		5.59	3.42	7	3.8%	0.4%
stops		5.48	3.37	7	3.8%	0.4%
slow		5.39	2.63	10	5.4%	0.9%
fast lane	✓	5.32	4.79	4	2.2%	0.1%
second pit	✓	4.96	4.56	4	2.2%	0.1%
come		4.93	1.74	17	9.2%	2.7%
coming pits	✓	4.88	3.88	5	2.7%	0.2%
60 kilometers	✓	4.86	5.36	3	1.6%	0.0%
high pressure	✓	4.86	5.36	3	1.6%	0.0%
pit crew	✓	4.86	5.36	3	1.6%	0.0%
slow pit	✓	4.86	5.36	3	1.6%	0.0%
started pit	✓	4.86	5.36	3	1.6%	0.0%
starting pit	✓	4.86	5.36	3	1.6%	0.0%
time pit	✓	4.86	5.36	3	1.6%	0.0%
make pit	✓	4.67	4.37	4	2.2%	0.1%
60		4.26	4.95	3	1.6%	0.1%
called pits	✓	4.26	4.95	3	1.6%	0.1%
exit pit	✓	4.26	4.95	3	1.6%	0.1%
pits comes	✓	4.26	4.95	3	1.6%	0.1%
starting		3.93	3.30	5	2.7%	0.3%
cheap		3.87	4.63	3	1.6%	0.1%
crews		3.87	4.63	3	1.6%	0.1%
called		3.85	3.79	4	2.2%	0.2%
crew		3.85	3.79	4	2.2%	0.2%
pit stops	✓	3.85	3.79	4	2.2%	0.2%
fast		3.64	3.11	5	2.7%	0.3%
going pit	✓	3.58	4.37	3	1.6%	0.1%
kilometers hour	✓	3.58	4.37	3	1.6%	0.1%
pits going	✓	3.58	4.37	3	1.6%	0.1%

C15 — Tires & compounds

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 83

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
tires		200.07	4.05	170	44.9%	2.7%
tire		106.09	4.01	96	25.3%	1.6%
tyres		67.12	4.13	58	15.3%	0.9%
hard		65.81	3.60	74	19.5%	1.6%
medium		51.97	3.97	49	12.9%	0.8%
compound		36.33	4.16	31	8.2%	0.5%
tyre		33.46	4.07	30	7.9%	0.5%
soft		31.02	3.86	31	8.2%	0.6%
fresher		26.27	3.80	27	7.1%	0.5%
compound tire	✓	22.33	4.33	17	4.5%	0.2%
fresher tires	✓	20.59	4.11	18	4.7%	0.3%
hard compound	✓	19.80	4.25	16	4.2%	0.2%
medium tires	✓	18.87	4.16	16	4.2%	0.2%
hard tires	✓	16.93	3.84	17	4.5%	0.3%
intermediate		16.80	3.94	16	4.2%	0.3%
temperature		16.18	3.99	15	4.0%	0.2%
hard tyres	✓	15.73	4.33	12	3.2%	0.2%
set		15.09	3.37	19	5.0%	0.5%
hard tire	✓	15.07	3.85	15	4.0%	0.3%
medium tire	✓	15.07	4.13	13	3.4%	0.2%
tires going	✓	12.56	4.09	11	2.9%	0.2%
soft tire	✓	12.08	4.20	10	2.6%	0.1%
soft tires	✓	11.36	3.89	11	2.9%	0.2%
intermediate tires	✓	11.32	4.07	10	2.6%	0.2%
rubber		10.80	3.15	15	4.0%	0.4%
hard tyre	✓	10.47	4.34	8	2.1%	0.1%
fresh		9.72	3.75	10	2.6%	0.2%
slick		9.72	3.75	10	2.6%	0.2%
traction		9.25	2.64	17	4.5%	0.7%
medium compound	✓	9.16	4.33	7	1.8%	0.1%
rear		8.46	2.25	20	5.3%	1.1%
new		8.17	2.63	15	4.0%	0.6%
medium tyre	✓	7.85	4.34	6	1.6%	0.1%
set tires	✓	7.85	4.34	6	1.6%	0.1%
compound tires	✓	7.02	4.11	6	1.6%	0.1%
fired		7.02	4.11	6	1.6%	0.1%
grip		6.69	2.21	16	4.2%	0.9%
pits		6.59	1.90	20	5.3%	1.4%
new set	✓	6.44	3.92	6	1.6%	0.1%
pirelli		6.44	3.92	6	1.6%	0.1%
set hard	✓	5.78	4.07	5	1.3%	0.1%
softs		5.78	4.07	5	1.3%	0.1%
better		5.30	1.96	15	4.0%	1.0%
cold		5.25	3.85	5	1.3%	0.1%

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
warm		5.25	3.85	5	1.3%	0.1%
12 laps	✓	5.23	4.34	4	1.1%	0.1%
compound tyre	✓	5.23	4.34	4	1.1%	0.1%
new tires	✓	5.23	4.34	4	1.1%	0.1%
rear axle	✓	5.23	4.34	4	1.1%	0.1%
tire temperature	✓	5.23	4.34	4	1.1%	0.1%
tires got	✓	5.23	4.34	4	1.1%	0.1%
tires lot	✓	5.23	4.34	4	1.1%	0.1%
younger		5.23	4.34	4	1.1%	0.1%
rid		4.99	3.33	6	1.6%	0.2%
window		4.74	3.22	6	1.6%	0.2%
axle		4.54	4.01	4	1.1%	0.1%
try end	✓	4.54	4.01	4	1.1%	0.1%
end		4.31	1.52	18	4.7%	1.7%
harder		4.08	3.75	4	1.1%	0.1%
soft compound	✓	4.08	3.75	4	1.1%	0.1%
tires just	✓	4.08	3.75	4	1.1%	0.1%
working		4.07	2.62	7	1.8%	0.3%
bring		3.94	2.56	7	1.8%	0.3%
bring tires	✓	3.92	4.33	3	0.8%	0.0%
fresh tires	✓	3.92	4.33	3	0.8%	0.0%
got tyres	✓	3.92	4.33	3	0.8%	0.0%
intermediate tyres	✓	3.92	4.33	3	0.8%	0.0%
laps younger	✓	3.92	4.33	3	0.8%	0.0%
temperature window	✓	3.92	4.33	3	0.8%	0.0%
tire grip	✓	3.92	4.33	3	0.8%	0.0%
tire medium	✓	3.92	4.33	3	0.8%	0.0%
tire run	✓	3.92	4.33	3	0.8%	0.0%
tire way	✓	3.92	4.33	3	0.8%	0.0%
tires gets	✓	3.92	4.33	3	0.8%	0.0%
tires getting	✓	3.92	4.33	3	0.8%	0.0%
tires make	✓	3.92	4.33	3	0.8%	0.0%
tires ve	✓	3.92	4.33	3	0.8%	0.0%
tyres fired	✓	3.92	4.33	3	0.8%	0.0%
tyres got	✓	3.92	4.33	3	0.8%	0.0%
warming		3.92	4.33	3	0.8%	0.0%
faster		3.76	3.07	5	1.3%	0.2%
fresher rubber	✓	3.76	3.07	5	1.3%	0.2%
mediums		3.57	2.96	5	1.3%	0.2%

C16 — Safety car deployment

Domain: Regulatory **Verdict:** Strict-core **Enriched terms:** 25

C16 — Safety car deployment: 25 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
safety		190.33	5.59	106	86.9%	1.8%
safety car	✓	189.72	5.62	105	86.1%	1.8%
car		129.58	4.09	109	89.3%	5.2%
virtual		58.43	5.81	34	27.9%	0.5%
virtual safety	✓	56.55	5.81	33	27.0%	0.5%
deployed		22.03	5.61	14	11.5%	0.2%
car deployed	✓	17.42	5.62	11	9.0%	0.2%
car safety	✓	12.65	5.97	7	5.7%	0.1%
ending		10.16	5.32	7	5.7%	0.1%
car ending	✓	9.02	5.97	5	4.1%	0.1%
going safety	✓	7.21	5.97	4	3.3%	0.1%
got safety	✓	7.21	5.97	4	3.3%	0.1%
lapped cars	✓	7.21	5.97	4	3.3%	0.1%
timing		6.65	4.97	5	4.1%	0.1%
car coming	✓	6.52	5.64	4	3.3%	0.1%
lapped		6.52	5.64	4	3.3%	0.1%
allowed overtake	✓	5.40	5.97	3	2.5%	0.0%
need safety	✓	5.40	5.97	3	2.5%	0.0%
barrier		5.03	3.55	6	4.9%	0.4%
car ll	✓	4.81	5.55	3	2.5%	0.1%
impact		4.81	5.55	3	2.5%	0.1%
end lap	✓	4.26	4.16	4	3.3%	0.2%
cars		4.17	2.53	8	6.6%	1.1%
surprised		3.88	4.74	3	2.5%	0.1%
barriers		3.68	3.16	5	4.1%	0.5%

C17 — Gap & pace

Domain: Strategy & technical **Verdict:** Moderate **Enriched terms:** 20

C17 — Gap & pace: 20 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
seconds		41.63	4.76	34	50.7%	1.9%
gap		26.50	4.82	22	32.8%	1.2%
tenths		10.21	5.03	8	11.9%	0.4%
hour		9.69	5.32	7	10.4%	0.3%
miles hour	✓	8.26	5.98	5	7.5%	0.1%
tenths second	✓	8.05	5.18	6	9.0%	0.2%
gap seconds	✓	6.73	6.04	4	6.0%	0.1%
miles		6.60	5.07	5	7.5%	0.2%
half		5.59	3.90	6	9.0%	0.6%
extra		5.43	4.35	5	7.5%	0.4%
20		4.93	4.75	4	6.0%	0.2%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
pace		4.55	3.31	6	9.0%	0.9%
10 seconds	✓	4.46	5.42	3	4.5%	0.1%
second		4.07	1.60	15	22.4%	7.4%
10		3.70	2.80	6	9.0%	1.3%
12 miles	✓	3.65	6.27	2	3.0%	0.0%
gap come	✓	3.65	6.27	2	3.0%	0.0%
gap tenths	✓	3.65	6.27	2	3.0%	0.0%
second gap	✓	3.65	6.27	2	3.0%	0.0%
tenths quicker	✓	3.65	6.27	2	3.0%	0.0%

C18 — Team dynamics

Domain: Narrative & meta **Verdict:** Robust **Enriched terms:** 5

C18 — Team dynamics: 5 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
teammate		27.82	2.24	65	10.8%	2.3%
team		16.57	2.26	38	6.3%	1.3%
ahead teammate	✓	6.29	3.09	8	1.3%	0.2%
ahead		4.42	0.78	56	9.3%	5.5%
past		3.52	0.73	48	8.0%	4.8%

C19 — Race lead holding

Domain: Race dynamics **Verdict:** Robust **Enriched terms:** 14

C19 — Race lead holding: 14 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
lead		64.46	3.80	68	59.1%	4.3%
leads		46.95	4.73	39	33.9%	1.3%
leads second	✓	7.42	5.38	5	4.3%	0.1%
lead second	✓	7.31	6.06	4	3.5%	0.1%
maintains lead	✓	7.07	5.21	5	4.3%	0.1%
holds lead	✓	6.78	5.06	5	4.3%	0.1%
lead comes	✓	6.62	5.74	4	3.5%	0.1%
holds		6.21	4.18	6	5.2%	0.3%
leads comes	✓	5.78	5.25	4	3.5%	0.1%
maintains		5.57	4.37	5	4.3%	0.2%
leads inside	✓	5.48	6.06	3	2.6%	0.0%
leads leads	✓	4.88	5.64	3	2.6%	0.1%
takes lead	✓	4.86	3.47	6	5.2%	0.5%
maintain lead	✓	3.95	4.84	3	2.6%	0.1%

C20 — Wing & car damage

Domain: Strategy & technical **Verdict:** Strict-core **Enriched terms:** 16

C20 — Wing & car damage: 16 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
wing		64.35	6.21	36	59.0%	0.8%
puncture		24.18	6.26	14	23.0%	0.3%
damage		21.42	5.24	16	26.2%	0.7%
damaged		16.82	6.55	9	14.8%	0.2%
got puncture	✓	8.82	6.29	5	8.2%	0.1%
contact		8.80	3.33	12	19.7%	2.0%
bit contact	✓	8.44	6.97	4	6.6%	0.1%
damage wing	✓	8.44	6.97	4	6.6%	0.1%
damaged wing	✓	8.44	6.97	4	6.6%	0.1%
rear		6.52	3.56	8	13.1%	1.1%
wing endplate	✓	6.32	6.97	3	4.9%	0.0%
think		6.19	2.72	11	18.0%	2.7%
endplate		5.72	6.55	3	4.9%	0.1%
ripped		5.72	6.55	3	4.9%	0.1%
rear wing	✓	5.21	4.97	4	6.6%	0.2%
wing damage	✓	3.73	6.38	2	3.3%	0.0%

C21 — Braking & lockups

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 19

C21 — Braking & lockups: 19 FDR-significant terms

Term	Bi	z -score	$\log_2$ FC	n	Cov%	Corp%
braking		72.31	5.18	51	39.8%	1.1%
brakes		43.39	4.47	39	30.5%	1.4%
braking zone	✓	27.51	5.18	20	15.6%	0.4%
zone		22.49	4.55	20	15.6%	0.7%
brake		16.72	4.96	13	10.2%	0.3%
late		12.62	3.26	18	14.1%	1.5%
late braking	✓	10.01	5.25	7	5.5%	0.1%
brake pedal	✓	8.92	5.90	5	3.9%	0.1%
pedal		8.14	5.64	5	3.9%	0.1%
lock		5.45	3.09	8	6.2%	0.7%
brakes just	✓	5.34	5.91	3	2.3%	0.0%
way traffic	✓	5.34	5.91	3	2.3%	0.0%
outside brakes	✓	4.74	5.48	3	2.3%	0.1%
curb		4.49	3.70	5	3.9%	0.3%
brakes late	✓	4.35	5.16	3	2.3%	0.1%
late brakes	✓	4.14	3.47	5	3.9%	0.4%
rear		4.09	2.49	8	6.2%	1.1%
super late	✓	4.05	4.91	3	2.3%	0.1%
zone turn	✓	4.05	4.91	3	2.3%	0.1%

C22 — Wheel-to-wheel**Domain:** Race dynamics **Verdict:** Strict-core **Enriched terms:** 11

C22 — Wheel-to-wheel: 11 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
wheel		110.01	5.64	64	90.1%	1.8%
wheel wheel	✓	88.81	5.93	51	71.8%	1.2%
going wheel	✓	26.22	5.99	16	22.5%	0.4%
wheel turn	✓	10.85	6.34	6	8.5%	0.1%
wheel goes	✓	6.12	6.74	3	4.2%	0.0%
goes wheel	✓	5.52	6.34	3	4.2%	0.1%
went wheel	✓	5.52	6.34	3	4.2%	0.1%
wheel action	✓	5.52	6.34	3	4.2%	0.1%
wheel gets	✓	5.52	6.34	3	4.2%	0.1%
wheels		5.22	4.22	5	7.0%	0.4%
went		4.45	2.92	7	9.9%	1.3%

C23 — Circuit action**Domain:** Race dynamics **Verdict:** Moderate **Enriched terms:** 4

C23 — Circuit action: 4 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
racetrack		109.79	6.16	58	80.6%	1.1%
corner racetrack	✓	9.42	6.48	5	6.9%	0.1%
turn racetrack	✓	6.97	6.16	4	5.6%	0.1%
turns racetrack	✓	3.58	6.16	2	2.8%	0.0%

C24 — Inter-car distance**Domain:** Strategy & technical **Verdict:** Fragile **Enriched terms:** 17

C24 — Inter-car distance: 17 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
car		40.14	3.74	42	70.0%	5.2%
cars		13.15	4.25	13	21.7%	1.1%
car length	✓	11.90	6.77	6	10.0%	0.1%
car lengths	✓	11.30	6.57	6	10.0%	0.1%
length		11.30	6.57	6	10.0%	0.1%
lengths		11.30	6.57	6	10.0%	0.1%
couple car	✓	7.77	6.68	4	6.7%	0.1%
width		7.29	6.41	4	6.7%	0.1%
got car	✓	6.38	5.82	4	6.7%	0.1%
length ahead	✓	6.34	6.97	3	5.0%	0.0%
car ahead	✓	5.79	5.41	4	6.7%	0.2%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
car width	✓	5.34	6.27	3	5.0%	0.1%
flat floor	✓	5.04	6.00	3	5.0%	0.1%
floor		4.14	5.12	3	5.0%	0.1%
flat		4.02	5.00	3	5.0%	0.2%
couple		3.90	3.34	5	8.3%	0.8%
positions car	✓	3.74	6.38	2	3.3%	0.0%

C25 — Driver & team references

Domain: Narrative & meta **Verdict:** Fragile **Enriched terms:** 8

C25 — Driver & team references: 8 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
driver		62.54	4.71	49	71.0%	2.7%
nation driver	✓	37.60	5.32	27	39.1%	1.0%
nation		32.21	4.61	28	40.6%	1.7%
drivers		13.68	4.38	13	18.8%	0.9%
driver getting	✓	6.15	6.80	3	4.3%	0.0%
board nation	✓	5.55	6.38	3	4.3%	0.1%
driver trying	✓	5.55	6.38	3	4.3%	0.1%
contract		3.62	6.21	2	2.9%	0.0%

C26 — Defending & attacking

Domain: Race dynamics **Verdict:** Moderate **Enriched terms:** 11

C26 — Defending & attacking: 11 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
fight		30.05	4.19	30	27.3%	1.5%
defend		24.69	5.05	19	17.3%	0.5%
battle		16.43	3.66	20	18.2%	1.4%
defensive		11.41	4.29	11	10.0%	0.5%
going fight	✓	8.81	5.38	6	5.5%	0.1%
try defend	✓	4.94	5.69	3	2.7%	0.1%
defended		4.55	5.38	3	2.7%	0.1%
try fight	✓	4.31	4.21	4	3.6%	0.2%
defense		3.70	3.73	4	3.6%	0.3%
fight		3.61	3.66	4	3.6%	0.3%
going		3.52	1.02	26	23.6%	11.7%

C27 — Overtake battles

Domain: Race dynamics **Verdict:** Robust **Enriched terms:** 17

C27 — Overtake battles: 17 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
inside		26.71	1.43	126	16.5%	6.1%
door		10.04	2.23	22	2.9%	0.6%
corner		7.47	1.15	50	6.6%	3.0%
gets away	✓	7.20	2.84	10	1.3%	0.2%
past		6.92	0.91	69	9.1%	4.8%
outside		6.65	0.93	64	8.4%	4.4%
open door	✓	6.45	3.01	8	1.0%	0.1%
getaway		5.44	2.40	10	1.3%	0.2%
space		5.13	1.64	18	2.4%	0.8%
ahead		4.64	0.71	68	8.9%	5.5%
close		4.03	0.93	36	4.7%	2.5%
goes corner	✓	4.01	3.33	4	0.5%	0.1%
leads corner	✓	4.01	3.33	4	0.5%	0.1%
past outside	✓	4.01	3.33	4	0.5%	0.1%
goes		3.92	0.64	67	8.8%	5.6%
inside goes	✓	3.53	2.23	7	0.9%	0.2%
inside line	✓	3.50	1.19	20	2.6%	1.2%

C28 — Corner battles

Domain: Race dynamics **Verdict:** Strict-core **Enriched terms:** 7

C28 — Corner battles: 7 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
turn		68.84	3.53	71	86.6%	7.5%
run turn	✓	10.39	5.54	7	8.5%	0.2%
abreast		6.38	5.74	4	4.9%	0.1%
way turn	✓	5.65	4.47	5	6.1%	0.3%
run		4.86	2.84	8	9.8%	1.4%
heading turn	✓	4.81	4.63	4	4.9%	0.2%
outside turn	✓	3.96	3.96	4	4.9%	0.3%

C29 — Corner specific moves

Domain: Race dynamics **Verdict:** Weak **Enriched terms:** 18

C29 — Corner specific moves: 18 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
turn		56.34	3.61	55	91.7%	7.5%
turn 11	✓	33.82	6.76	17	28.3%	0.3%
11		28.82	6.18	17	28.3%	0.4%
turn 10	✓	13.34	6.62	7	11.7%	0.1%
turn seven	✓	11.30	6.57	6	10.0%	0.1%

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
13		7.32	5.51	5	8.3%	0.2%
turn 12	✓	6.99	5.31	5	8.3%	0.2%
12		6.61	4.45	6	10.0%	0.5%
inside turn	✓	6.57	5.07	5	8.3%	0.2%
seven		5.94	4.09	6	10.0%	0.6%
heading turn	✓	5.36	5.08	4	6.7%	0.2%
heading		5.27	3.72	6	10.0%	0.8%
coming turn	✓	5.24	4.99	4	6.7%	0.2%
turn 13	✓	5.04	6.00	3	5.0%	0.1%
10		4.99	3.19	7	11.7%	1.3%
14		4.82	4.67	4	6.7%	0.3%
turn 14	✓	4.02	5.00	3	5.0%	0.2%
away turn	✓	3.74	6.38	2	3.3%	0.0%

C30 — Lap pace & timing

Domain: Strategy & technical **Verdict:** Robust **Enriched terms:** 23

C30 — Lap pace & timing: 23 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
lap		196.75	4.55	135	99.3%	4.2%
fastest lap	✓	31.10	4.84	25	18.4%	0.6%
fastest		26.22	4.33	25	18.4%	0.9%
opening		14.18	4.02	15	11.0%	0.7%
previous lap	✓	14.09	5.82	8	5.9%	0.1%
opening lap	✓	13.94	4.88	11	8.1%	0.3%
previous		11.89	5.35	8	5.9%	0.1%
lap lap	✓	9.71	5.60	6	4.4%	0.1%
turn lap	✓	7.47	5.33	5	3.7%	0.1%
lap goes	✓	5.85	5.22	4	2.9%	0.1%
lap going	✓	5.85	5.22	4	2.9%	0.1%
lap time	✓	5.85	5.22	4	2.9%	0.1%
quicker		5.65	3.87	6	4.4%	0.3%
end lap	✓	5.54	4.33	5	3.7%	0.2%
middle		4.67	3.35	6	4.4%	0.4%
corners lap	✓	4.66	5.40	3	2.2%	0.1%
sector lap	✓	4.66	5.40	3	2.2%	0.1%
seconds		4.36	2.11	11	8.1%	1.9%
middle sector	✓	4.22	4.11	4	2.9%	0.2%
lap number	✓	3.98	4.81	3	2.2%	0.1%
sector		3.93	3.33	5	3.7%	0.4%
point fastest	✓	3.74	4.60	3	2.2%	0.1%
seconds away	✓	3.54	4.40	3	2.2%	0.1%

C31 — Race leader changes**Domain:** Race dynamics **Verdict:** Strict-core **Enriched terms:** 18

C31 — Race leader changes: 18 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
race		192.43	3.96	157	96.9%	6.2%
lead race	✓	39.33	5.12	28	17.3%	0.5%
lead		21.59	2.61	42	25.9%	4.2%
race leader	✓	10.07	4.13	10	6.2%	0.4%
leader		9.83	3.40	13	8.0%	0.8%
takes lead	✓	7.40	3.56	9	5.6%	0.5%
race lead	✓	6.08	4.06	6	3.7%	0.2%
start race	✓	5.78	4.42	5	3.1%	0.1%
started race	✓	5.55	4.98	4	2.5%	0.1%
win race	✓	4.65	4.39	4	2.5%	0.1%
going lead	✓	4.43	4.24	4	2.5%	0.1%
race going	✓	4.43	4.24	4	2.5%	0.1%
race control	✓	4.24	4.10	4	2.5%	0.1%
lap race	✓	4.21	3.49	5	3.1%	0.3%
race far	✓	4.04	4.82	3	1.9%	0.1%
review		4.04	4.82	3	1.9%	0.1%
place race	✓	3.66	3.65	4	2.5%	0.2%
retake lead	✓	3.51	4.34	3	1.9%	0.1%

C32 — Front position changes**Domain:** Race dynamics **Verdict:** Fragile **Enriched terms:** 7

C32 — Front position changes: 7 FDR-significant terms

Term	Bi	<i>z</i> -score	log ₂ FC	<i>n</i>	Cov%	Corp%
fourth		14.30	4.02	15	33.3%	2.1%
place		13.36	2.65	23	51.1%	8.1%
position		8.73	2.75	15	33.3%	5.0%
fourth place	✓	7.84	4.54	7	15.6%	0.7%
second		4.85	1.96	13	28.9%	7.4%
second place	✓	4.53	2.94	7	15.6%	2.0%
couple positions	✓	3.70	6.38	2	4.4%	0.1%

C33 — Field position changes**Domain:** Race dynamics **Verdict:** Robust **Enriched terms:** 16

Term	Bi	z-score	$\log_2$ FC	n	Cov%	Corp%
seventh		31.45	4.66	27	22.1%	0.9%
place		27.81	2.47	55	45.1%	8.1%
sixth		24.11	4.22	24	19.7%	1.1%
fifth		16.02	3.73	19	15.6%	1.2%
sixth place	✓	14.90	4.85	12	9.8%	0.3%
eighth		14.32	3.88	16	13.1%	0.9%
seventh place	✓	12.65	4.90	10	8.2%	0.3%
11th		11.50	4.59	10	8.2%	0.3%
position		11.32	2.21	28	23.0%	5.0%
fifth place	✓	8.73	4.05	9	7.4%	0.4%
sixth position	✓	8.54	4.33	8	6.6%	0.3%
10th		7.97	4.11	8	6.6%	0.4%
ninth		5.66	3.19	8	6.6%	0.7%
12th		5.44	4.29	5	4.1%	0.2%
seventh position	✓	4.40	4.27	4	3.3%	0.2%
11th place	✓	3.51	4.38	3	2.5%	0.1%